Bushra Alzadjali



Ghaf Root
AGENDA

Ghaf Root Team

Lessons Learned

Al Ghaf Root

Approach & Tools

Green Root Story

Impact & Value

Closing



BUSHRA ALZADJALI

Ghaf Root TEAM



Bushra Alzadjali

- Data Science Engineer @ Space42
- NSA program x2 Graduate | EO & SMSE
- Space Entrepreneurship Master Graduate
- Green Youth Majlis Member
- ESGRS affiliate
- Aerospace Engineering Graduate
- Nomu Al Ghurair Women in AI Graduate
- Space Hackathon for Sustainability Participant

Hackathon Tips

Lessons from Ghaf Root, sharpened through the National Space Academy (NSA) x Space42 Earth Observations program

01

Pick a problem you can actually build

Passion matters, but in a hackathon, technical feasibility inside the time limit matters just as much. Choose a problem you have a real shot at solving.

02

Ask questions — constantly

The more you ask, the more you learn. A quick question could save you hours.

03

Apply what you learn

Test what you study right away. Applying it fast shows you exactly where your real strengths and weaknesses are.

04

Validate early and fast

Confirm the problem is real before you build the solution. This is the habit NSA's program with Space42 sharpened in me the most.

05

Aim higher

Build to more than just to finish — other teams are pushing for something unique too. “Shoot for the moon. Even if you miss, you'll land among the stars.”

This mindset carried forward: National Space Academy graduate (2 programs) → now Data Science Engineer at Space42

Team Collaboration

Ghaf Root was a solo project on paper.

01

Know each other's strengths

Understanding what each teammate is best at speeds up delivery — and in the long run, everyone ends up learning something new from each other.

02

Keep everyone in the loop

Even across separate teams, we shared the issues we ran into instead of everyone solving the same problem alone — that saved real time for all of us.

03

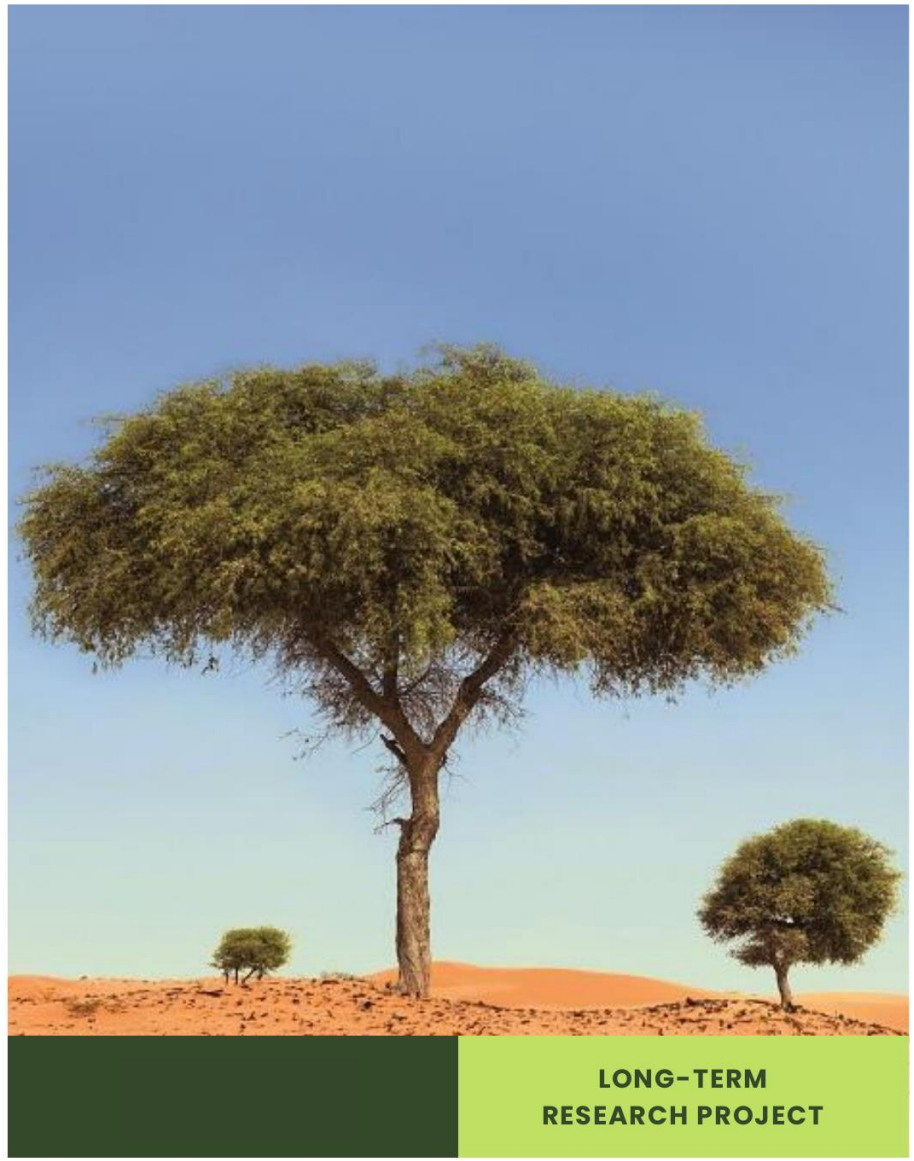
Escalate smart

Try it yourself → search for it → ask colleagues → ask mentors.
Move through each step quickly instead of getting stuck on one.

Though registered as a solo participant, real progress came from constant exchange with fellow participants and their teams.

AI-Ghaf Root

The UAE National Tree Detection through the *Eyes* of Satellites



Why GR as a solution?

What is GR’s impact?

- **Challenge**
  - Un-available, inefficient, and unscalable monitoring
- **Significance**
  - National, environmental, and cultural icon of the UAE
- **Impact**
  - Enables scalable, data-driven ecological monitoring
  - Supports urban greening & smart environmental planning
  - Aligns with UAE's sustainability and innovation goals
- **Solution**
  - AI + satellite imagery for automated tree detection











Solution Overview

NDVI | Normalized Difference Vegetation Index*

1.Ghaf Detection
·AI-based tree detection
·Barcode integration

2.Ghaf Monitoring
·Green area estimation
·NDVI* based health checks
·Parasite and stress detection

3.Environmental Modeling
·Ecosystem trends tracking
·Carbon modeling

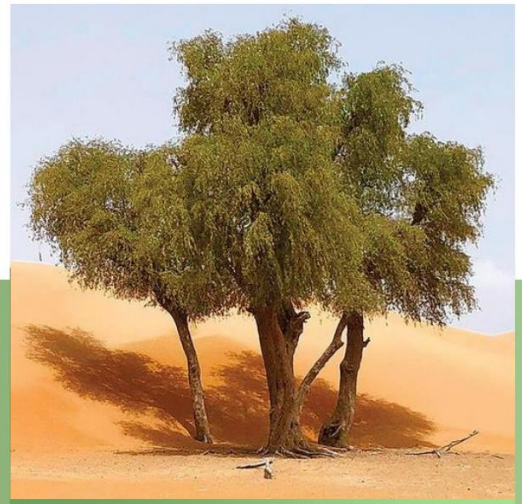
4.Tree Lifecycle & Insights
·AI-based age estimation
·Growth and anomaly tracking

5. Data & Platform Integration
·Live on gIQ platform (future-ready)
·API for external systems

GreenScope



Palm



Ghaf



Mangrove

GHAF ROOT

BUSHRA ALZADJALI



DATA & TOOLS

Ghaf Root

01

DATA

- SAR
- Multispectral

02

TOOLS

- GIQ Platform
- AI & ML Models
- Python language

03

Application Steps

- Data collection and preprocessing
- Manual annotation
- Model training and iterative tuning
- Model testing and validation
- Results analysis and reporting



Green Approach



01

Short Term Project

- Palm Tree Detection with Optical Images.
- 100% Completed.

02

Ghaf Root Project

- A Space Based Earth Observation Application.
- Including the detection of Ghaf & Mangrove trees.
- UAE National Trees and enhancing its ecological system.

GREEN ROOT by Bushra Alzadjali

- UAE
- Abu Dhabi
- PALM 01
- PALM 02
- Al Ain
- PALM 03
- PALM 04
- GHAF 01

UAE

"The Ghaf tree was declared the national tree of the UAE in 2008 because of its cultural and traditional significance." EAD

GIQ SPACE42

GREEN ROOT by Bushra Alzadjali

- UAE
- Abu Dhabi
- PALM 01
- PALM 02
- Al Ain
- PALM 03
- PALM 04
- GHAF 01

Abu Dhabi

"Ghaf trees can live for up to 120 years on average."
EAD

GIQ SPACE42

GREEN ROOT by Bushra Alzadjali



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- GHAF 01

PALM 01

"UAE, UN body join forces to eradicate red palm weevil in the region" GULF NEWS



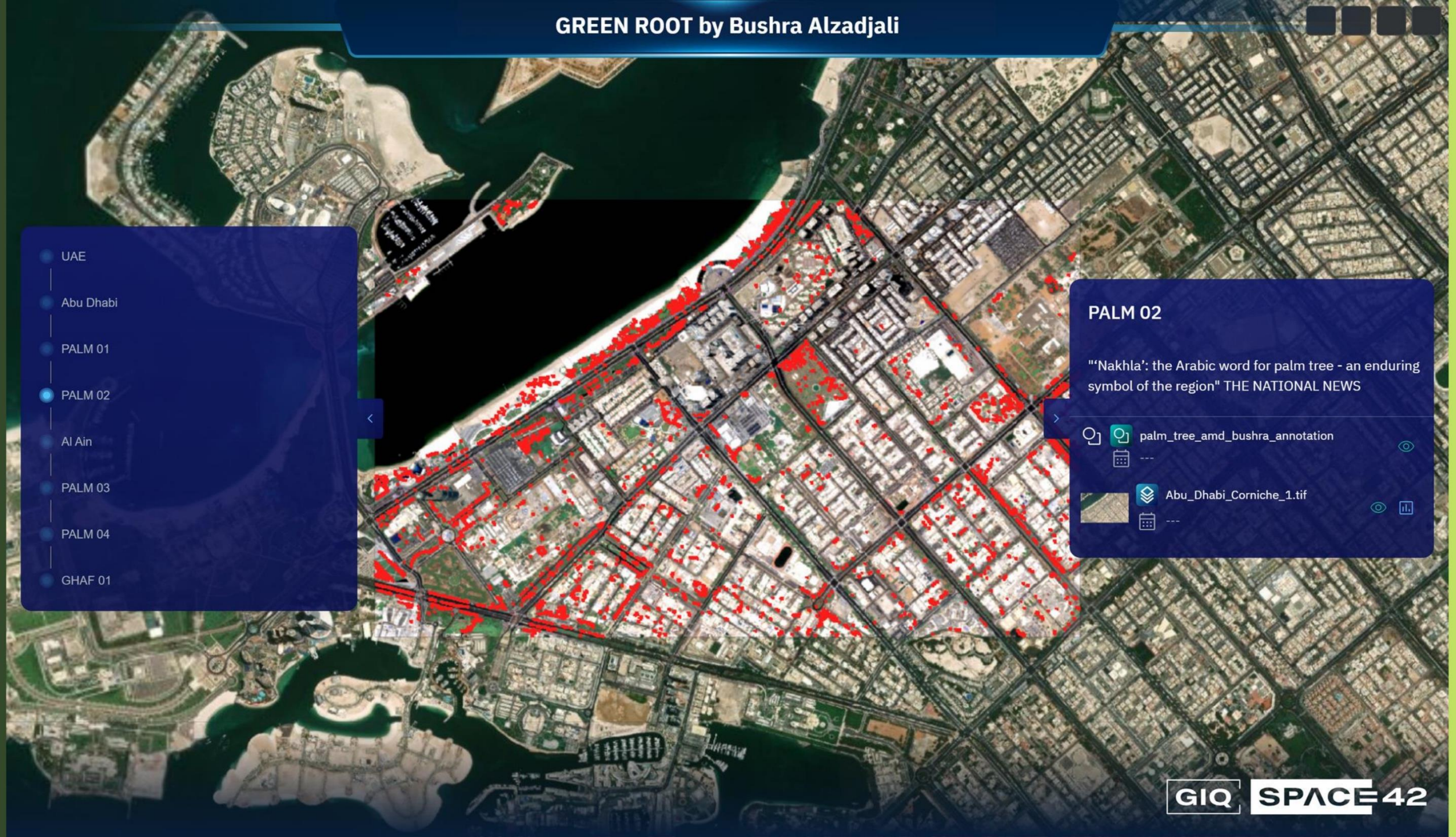
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12/May/2025

palm_tree_amd_bushra_annotation

GIQ SPACE42

GREEN ROOT by Bushra Alzadjali



GREEN ROOT by Bushra Alzadjali

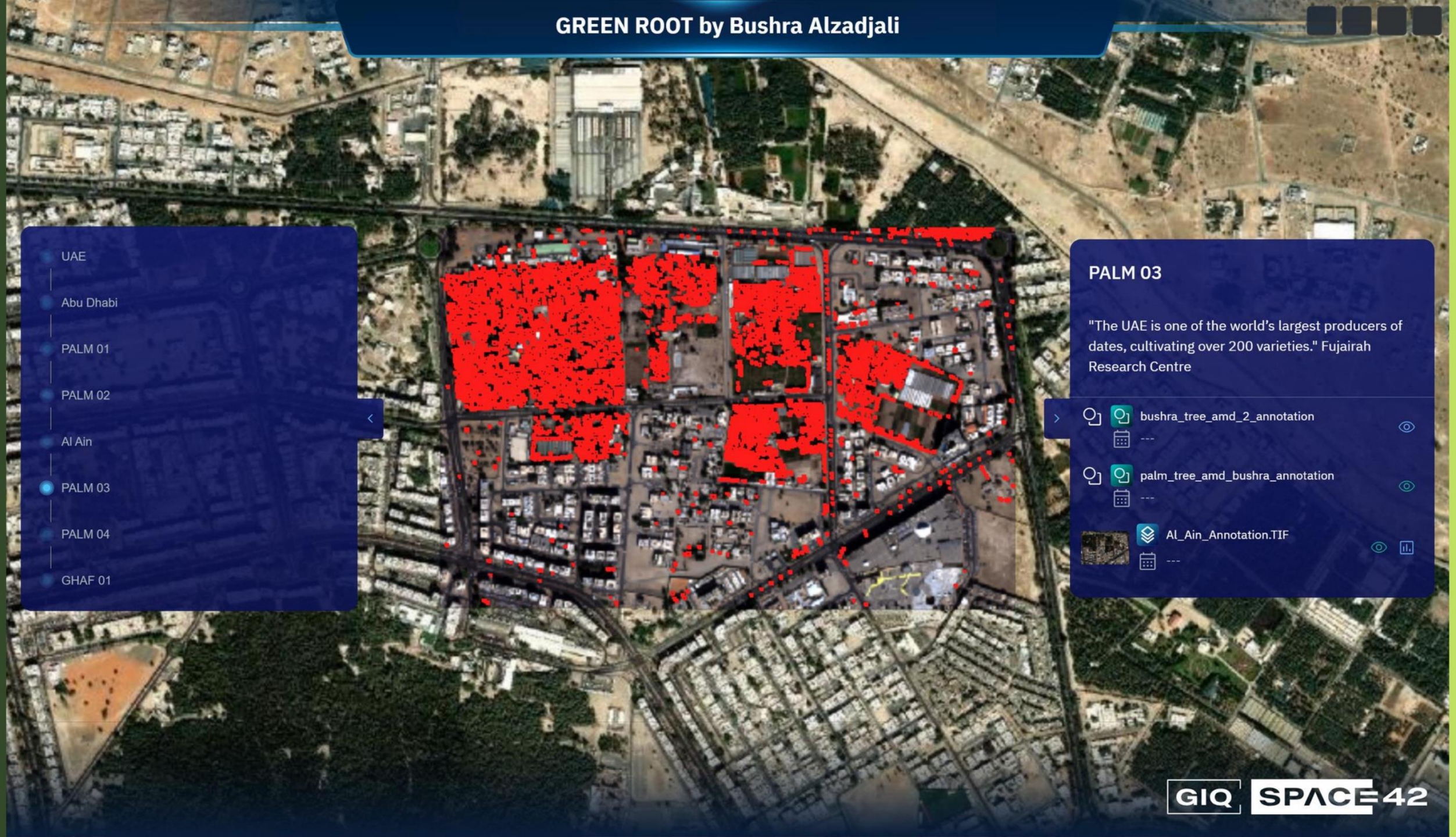
- UAE
- Abu Dhabi
- PALM 01
- PALM 02
- Al Ain
- PALM 03
- PALM 04
- GHAF 01

Al Ain

"According to UAE Federal Law No. 24 of 1999, it is prohibited to cut wild plants, uproot them or collect them illegally." EAD

GIQ SPACE42

GREEN ROOT by Bushra Alzadjali



- UAE
- Abu Dhabi
- PALM 01
- PALM 02
- Al Ain
- PALM 03
- PALM 04
- GHAF 01

PALM 03

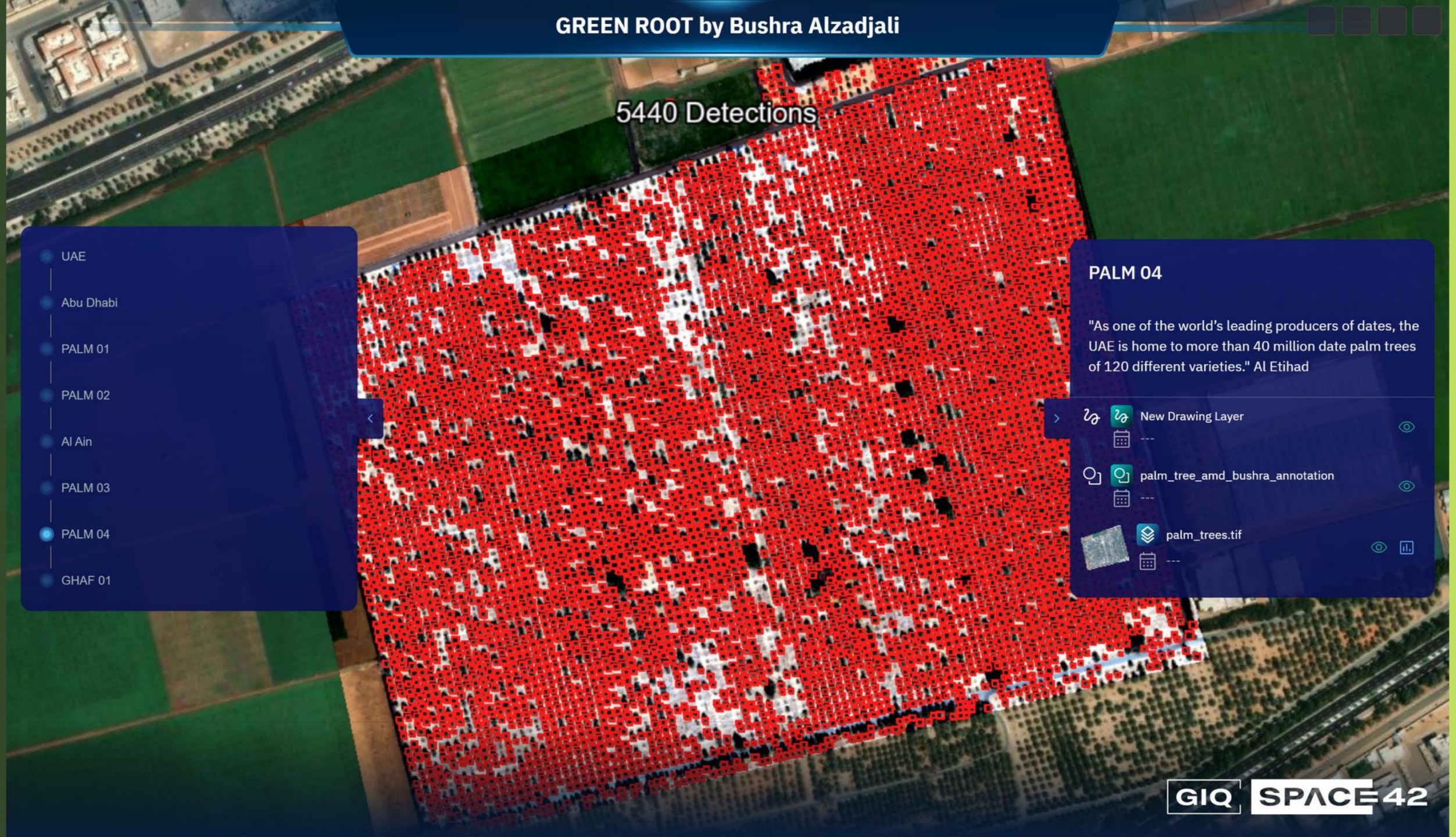
"The UAE is one of the world's largest producers of dates, cultivating over 200 varieties." Fujairah Research Centre

bushra_tree_amd_2_annotation
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palm_tree_amd_bushra_annotation

Al_Ain_Annotation.TIF

GIQ SPACE42


GREEN ROOT by Bushra Alzadjali
5440 Detections
UAE
Abu Dhabi
PALM 01
PALM 02
Al Ain
PALM 03
PALM 04
GHAF 01
PALM 04
"As one of the world's leading producers of dates, the UAE is home to more than 40 million date palm trees of 120 different varieties." Al Etihad
New Drawing Layer
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palm_trees.tif

GIQ
SPACE42

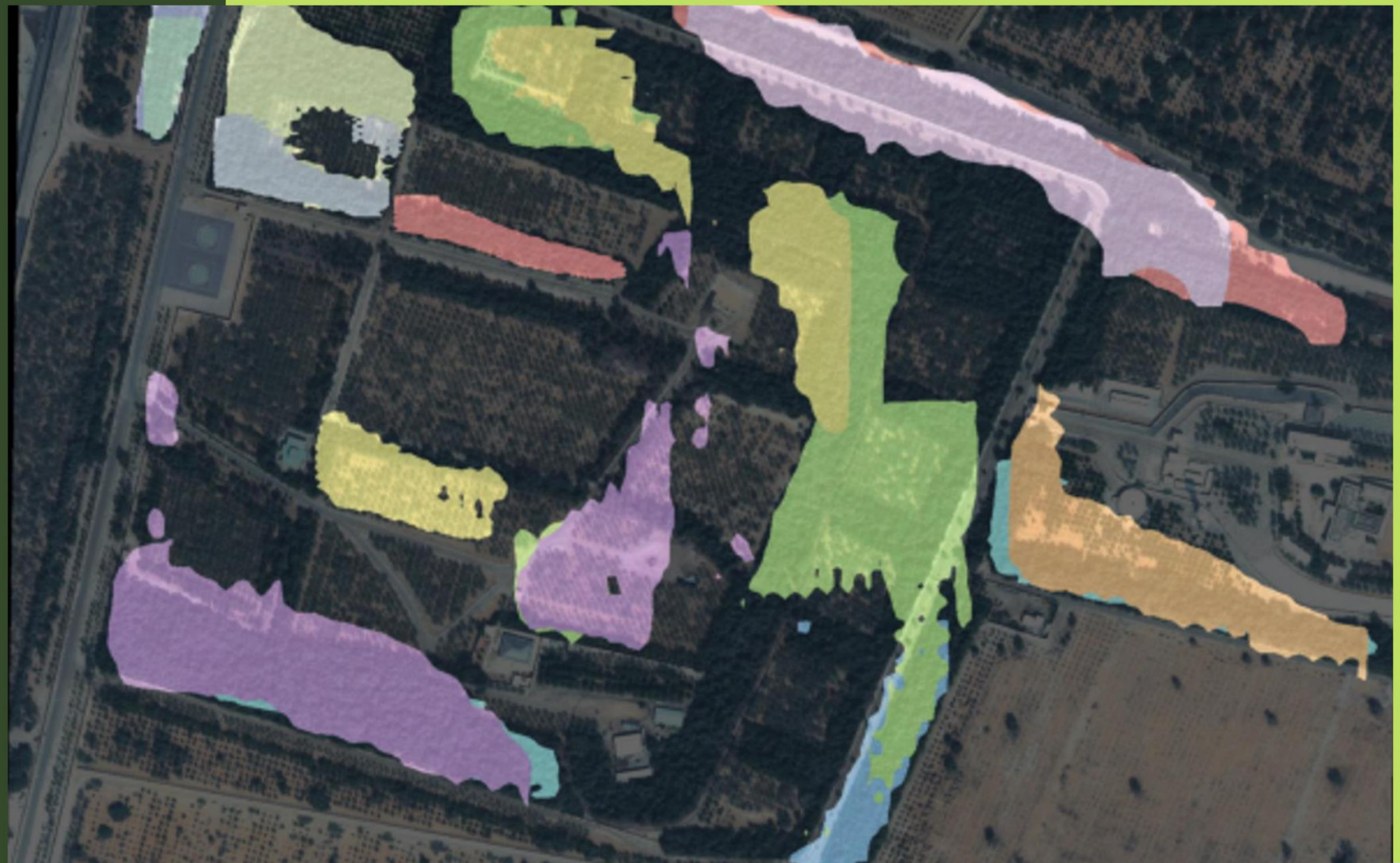
GREEN ROOT by Bushra Alzadjali



GHAF Detection

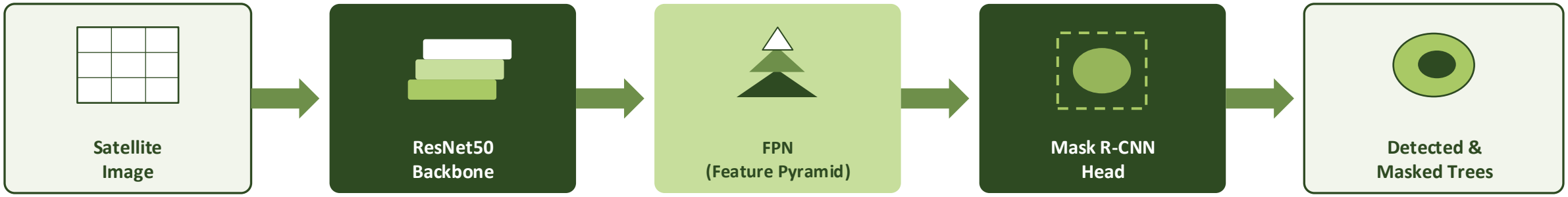
Segmentation Model

maskrcnn_resnet50_fpn



Model Architecture

maskrcnn_resnet50_fpn — how an image becomes tree masks



01 Mask R-CNN

The framework that detects each tree and outputs a precise pixel mask of its canopy, not just a box.

02 ResNet50

The 50-layer backbone that extracts visual features from the image using deep residual connections.

03 FPN

Combines features at multiple scales so both large clusters and small individual trees are caught.

Task type: instance segmentation | Model class: convolutional neural network (CNN) | Learning: supervised

Why It Fits

Matching the model to what Ghaf tree detection actually demands

01

Irregular, overlapping canopies

Ghaf trees cluster and their canopies merge visually from above. Instance segmentation separates each tree instead of returning one blended blob.

02

Wide range of tree sizes

From young saplings to large mature canopies — FPN's multi-scale detection catches both in the same pass.

03

Precise area, not just presence

Pixel-level masks (not boxes) allow accurate canopy area measurement — directly feeding green-area estimation and NDVI health checks.

04

Proven on aerial & satellite imagery

ResNet50 backbones are widely validated on remote sensing imagery, making the model reliable on SAR and multispectral inputs.

Known Tradeoffs

Being upfront about the cost of this accuracy

01

Needs labeled data

Accuracy depends directly on the volume and quality of manual annotation — the same step covered in Data & Tools.

02

Heavier compute

The deeper ResNet50 backbone means slower training and inference than lightweight detectors; not built for real-time use.

03

Small or occluded trees are harder

Very small saplings or heavily shadowed trees can be missed or merged into neighboring masks.

Mitigation: iterative retraining as more trees are annotated, and tuning anchor sizes to catch smaller canopies.

Ghaf Root

وكالة الإمارات للفضاء
UAE SPACE AGENCY

وزارة التغير المناخي والبيئة
MINISTRY OF CLIMATE CHANGE & ENVIRONMENT

هيئة البيئة - أبوظبي
Environment Agency - ABU DHABI

SPACE42

مركز محمد بن راشد للفضاء
MOHAMMED BIN RASHID SPACE CENTRE

بلدية مدينة أبوظبي
ABU DHABI CITY MUNICIPALITY

ESCRS
الجمعية الإماراتية لنظم المعلومات الجغرافية والاستشعار عن بعد
The Emirati Society of GIS and Remote Sensing

MOHAMED BIN ZAYED UNIVERSITY OF ARTIFICIAL INTELLIGENCE LIBRARY

بلدية دبي
Dubai Municipality

Who Benefits



BUSHRA ALZADJALI

- Government Agencies
- Environmental Organizations
- Urban Planners & Smart City Developers
- Academia & Research Institutions
- Tech & Satellite Data Providers
- Educational & Cultural Entities
- Agricultural & Forestry Departments

BUSHRA ALZADJALI



Solution Value

Environmental Stewardship
- Preserves Al-*Ghaf* trees

Advanced Earth Observation
- AI-Scalable monitoring

Strategic National Alignment
- *UAE Vision 2031 | Net Zero 2050 | SDG*

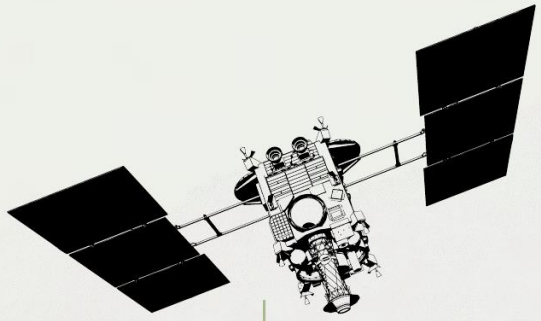
Efficient Resource Management
- Cuts *time* and *cost*

نحن الإمارات
WE THE UAE 2031

SUSTAINABLE DEVELOPMENT GOALS

Ghaf Root

Bushra Alzadjali



Space-Powered Sustainability

Ghaf Root | Ghaf Root | Ghaf Root | Ghaf Root | Ghaf Root | Ghaf Root | Ghaf Root | Ghaf Root | Ghaf Root | Ghaf Root

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